

Qlue: Boosting intelligence about activity in Indonesian cities

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With Google Cloud, Qlue delivers a range of Smart City applications that improve the lives of residents in Indonesian cities.

[About Qlue](#)



About Qlue

PT. Qlue Performa Indonesia (Qlue) aims to deliver Smart City technologies to cities across the country and in other Southeast Asian markets.

Google Cloud results

- Enabled city officials to respond to incidents in

500K citizens
90K users
incidents
daily

Industries: Technology

Location: Indonesia

minutes to
hours,
improving
the lives of
city
residents
in one of
the world's
most
diverse
countries

- Established
a platform
to expand
into several
new cities
across
Indonesia
- Maintained
a lean
technology
team
despite
rapid
growth

Compute Engine
(<https://cloud.google.com/compute/>)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

App Engine (<https://cloud.google.com/appengine/>)

Cloud SQL (<https://cloud.google.com/sql/>)

One of the world's largest, most diverse and vibrant cities, Jakarta faces a range of challenges delivering a safe, orderly environment for its more than 10 million residents. In 2014, administrators launched a Smart City program that aimed to use a range of technologies to improve quality of life in the Indonesian capital. Established to provide many of the applications and services driving the program in

Indonesia, PT. Qlue Performa Indonesia (Qlue).

(<https://qlue.co.id/site/>) now aims to deliver Smart City technologies to cities across the country and in other Southeast Asian markets.

Qlue applications include a city dashboard that enables city officials and service providers to track and resolve incidents of flooding, crime, fire, illegal rubbish dumping, and more. It monitors energy usage and traffic congestion in real time. Additional Qlue applications include a civic engagement app available on Google Play and the App Store that enables citizens to report incidents, and two enterprise-based applications—a workplace collaboration tool enhanced with geotagging information, and a billing and resident report management tool for property developers.

According to COO, Surya Darmadi, Qlue measures its success by its performance in transforming cities around Indonesia and beyond to become more effective and efficient. “We are introducing Qlue to major cities, related government agencies, and property developers,” he says.

As a provider of web and mobile applications, Qlue relies on its technology strategy to drive the success of its business. The organisation’s key technology projects power the collection, processing, and visualisation of data collected through social tools and sensors to make cities more effective and efficient.

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—Surya Darmadi, Chief
Operating Officer, Qlue

Qlue needed a platform to help run these functions and scale to support increasing loads on its web services and database systems due to expansion into more domestic and international cities. The need to scale without investing up to US\$200,000 up-front ruled out the purchase of servers, storage, networking, and associated systems. The cloud was the only option.

Qlue evaluated a range of services before deciding to run its applications on Google Cloud (<https://cloud.google.com/>). The organisation's assessment found using Google Cloud would remove the need for internal team members to undertake time and resource intensive administration tasks. "If we used a conventional infrastructure-as-a-service offering, we would have had to dedicate a team to manage operating systems, libraries, services, and load balancing," Darmadi says. "With Google Cloud, we are able to focus purely on developing our applications and growing the business."

Furthermore, the technical team at Qlue took tutorials and reviewed documentation to build confidence in using Google Cloud. "Our technical experts understood how to run scripts to create activities involving Google Cloud services better than in any other cloud service," says Darmadi. "Furthermore, they had experience and insights into how to build a scalable system in Google Compute Engine (<https://cloud.google.com/compute/>) and Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine/>)."

In December 2014, Qlue released the Qlue 1.0 civic engagement app to the general public. Focused on handling complaints with fewer interactions among users, the app runs on Google Cloud. The organisation's Google Cloud architecture now comprises Google App Engine

(<https://cloud.google.com/appengine/>) to deliver a fully managed platform for building web and mobile applications; Google Compute Engine to provide virtual machines with persistent disk storage in a scalable, load-balanced environment; and Google Cloud SQL (<https://cloud.google.com/sql/docs/>) to manage a MySQL database.

With the architecture stable and meeting key business requirements such as scalability, high levels of uptime, and the ability to back up and recover data, Qlue is planning to upgrade to Google Kubernetes Engine to further automate the scaling of technology resources. "We like the Google Kubernetes Engine capability to provide automated scaling as part of its cluster management functionality, as well as orchestration for running our Docker containers," says Darmadi. "Once we build our expertise in the Kubernetes product that powers Google Kubernetes Engine, we expect Google Kubernetes Engine to be the cornerstone of our Google Cloud architecture."

Google Cloud is now pervasive throughout the Qlue business, with several teams and individuals accountable for business activities that rely on Google Cloud. These include a quality assurance team that monitors and tests the web services API as well as mobile and web applications; a backend

team that builds, manages, and deploys the web services API; a mobile and web frontend team that builds and manages the Qlue mobile and web applications; and a database administrator who maintains the Qlue database.

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Lean tech team building apps

“Standardising on Google Cloud has met our expectations of keeping our information technology team lean and focused on deploying applications,” says Darmadi. “In addition, we can provide updates and new features more quickly with minimal downtime.”

Qlue is using an API to communicate data with its Android, iOS, and web applications, and is recording latency of less than 1,000 milliseconds. This helps ensure its mobile applications can respond quickly when accessing data from its servers.

Furthermore, using Google App Engine has enabled Qlue to build its expertise in configurations for scaling systems and architectures within Google Cloud. “This enables us to focus on developing and optimising applications,” says Darmadi. Qlue has now been downloaded 500,000 times and has 90,000 active users submitting an average of 4,000 reports per day.

Qlue is now focusing its operations in Jakarta, Manado, Kota Bima, and Probolinggo. The business is also collaborating with a range of agencies including Jakarta’s police (POLDA), The National Board for Disaster Management (BNPB), and The National Board for Narcotics Prevention (BNN).

Go-to platform for citizens' voices

"Qlue has become the go-to platform for citizens to voice their aspirations," says Darmadi. "The city can respond to complaints within hours, even minutes in cases such as traffic jams and trash-related reports."

Using Qlue mapping technology, in 2016 Jakarta reduced the number of areas flooded during monsoon season from 2,000 to 88. Qlue also helped reduce sanitation problems by 18% and cut the operating costs of Jakarta services by 35%. Citizens are now 27% more satisfied with the performance of Jakarta officials than before the technology was deployed.

Qlue regards Google Cloud as integral to its corporate culture. "As an IT company that builds and provides various mobile and web applications, cloud services are an extremely important part of our solution," says Darmadi.